

Mobile Computing – Satellite Systems

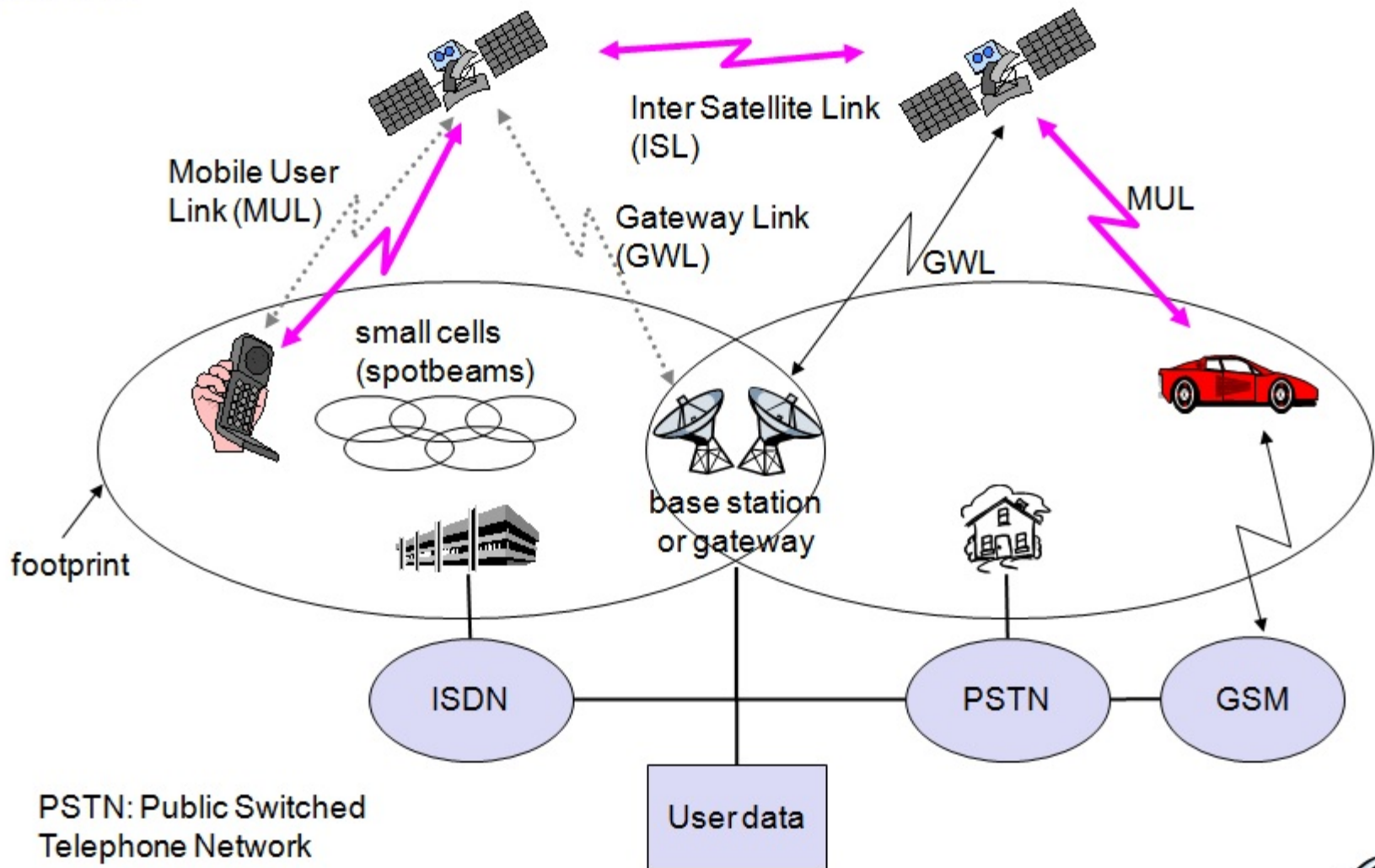
- 1945 Arthur C. Clarke publishes an essay about “Extra Terrestrial Relays”
- 1957 First satellite SPUTNIK
- 1960 First reflecting communication satellite ECHO
- 1963 First geostationary satellite SYNCOM
- 1965 First commercial geostationary satellite Satellite “Early Bird” (INTELSAT I): 240 duplex telephone channels or 1 TV channel, 1.5 years lifetime
- 1976 Three MARISAT satellites for maritime communication
- 1982 First **mobile satellite telephone system** INMARSAT-A
- 1988 First satellite system for **mobile phones and data communication** INMARSAT-C
- 1993 First **digital satellite telephone system**
- 1998 Global satellite systems for **small mobile phones**

Applications

- Traditionally
 - **Weather** satellites
 - **Radio and TV broadcast** satellites
 - **Military** satellites
 - Satellites for **navigation and localization** (e.g G.P.S)
- Telecommunication – Mobile
 - **Global telephone** backbone
 - Connections for communication in **remote places/under developed areas**
 - **Global mobile communication**

Classical satellite systems

Universal Mobile

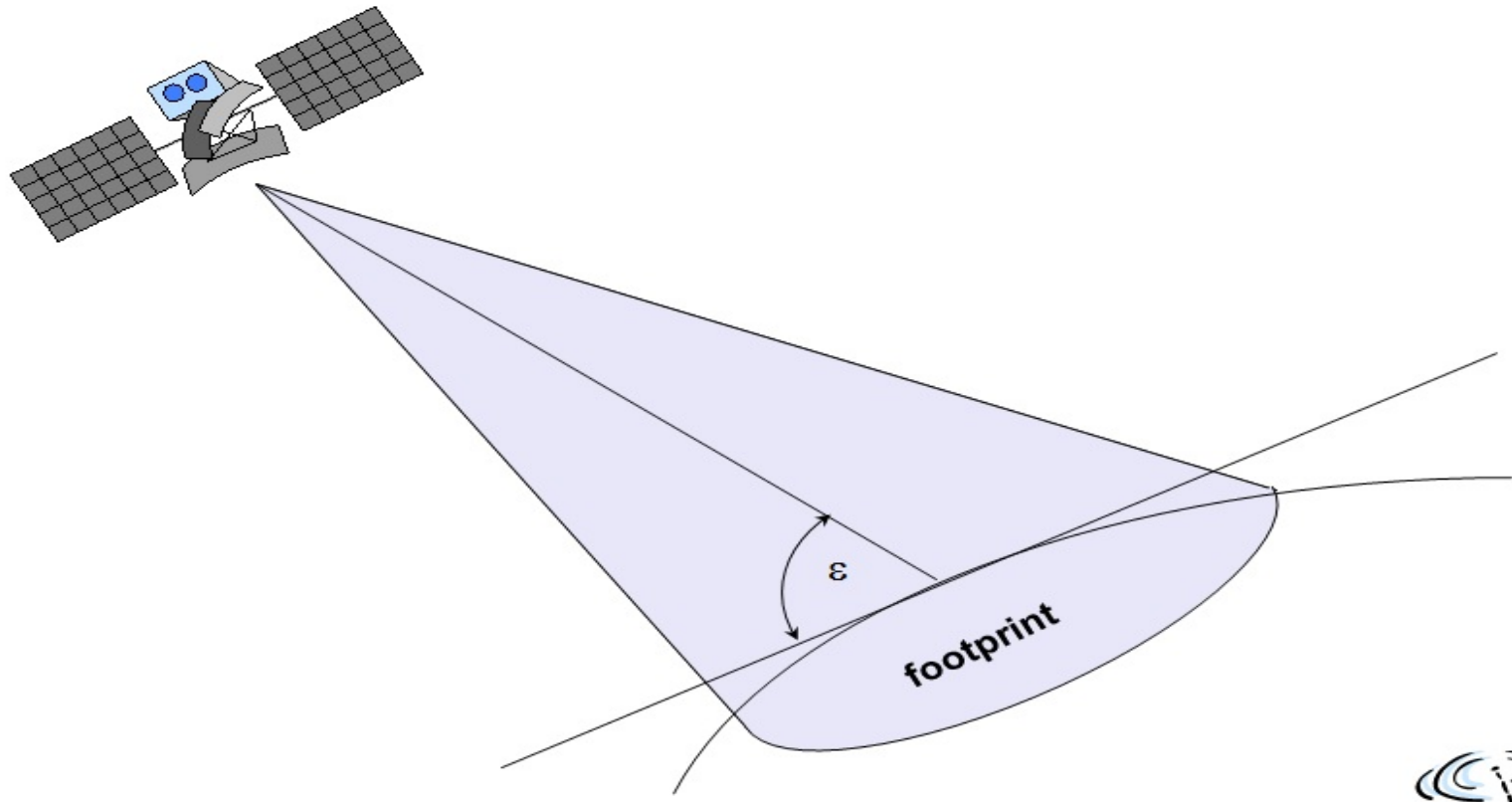


Basics of satellite Systems

- **Elliptical or circular** orbits
- Complete **rotation time** depends on **distance satellite-earth**
- **Inclination**: angle between orbit and equator
- **Elevation**: angle between satellite and horizon
- **LOS** (Line of Sight) to the satellite necessary for connection
 - High elevation needed, less absorption due to e.g. buildings
- **Uplink**: connection base station - **satellite**
- **Downlink**: connection satellite - **base station**
- Typically separated frequencies for uplink and downlink
 - **Transponder** used for sending/receiving and shifting of **frequencies**

Elevation

- **Elevation:** Angle between center of satellite beam and surface
- **Minimal Elevation:** Elevation needed atleast to communicate with the satellite



Factors Determining Satellite Systems

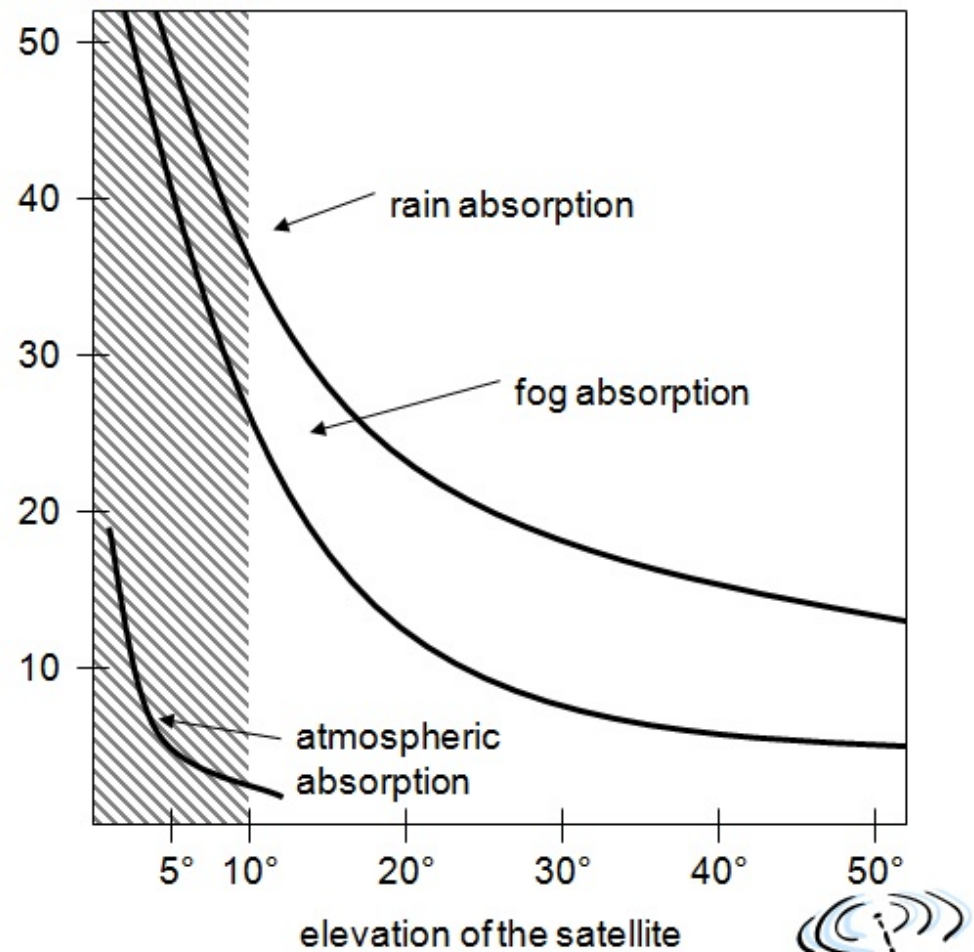
- **Parameters** like **attenuation** or received **power** determined by four parameters
 - **Sending power**
 - **Spreading**
 - **Gain of sending antenna**
 - **Distance** between sender and receiver
 - Based on gateway
 - **Gain of receiving antenna**
- **Problems**
 - Varying strength of received signal due to **multipath propagation**
 - Interruptions due to shadowing of signal (**no LOS**)

Atmospheric attenuation



Attenuation of
the signal in %

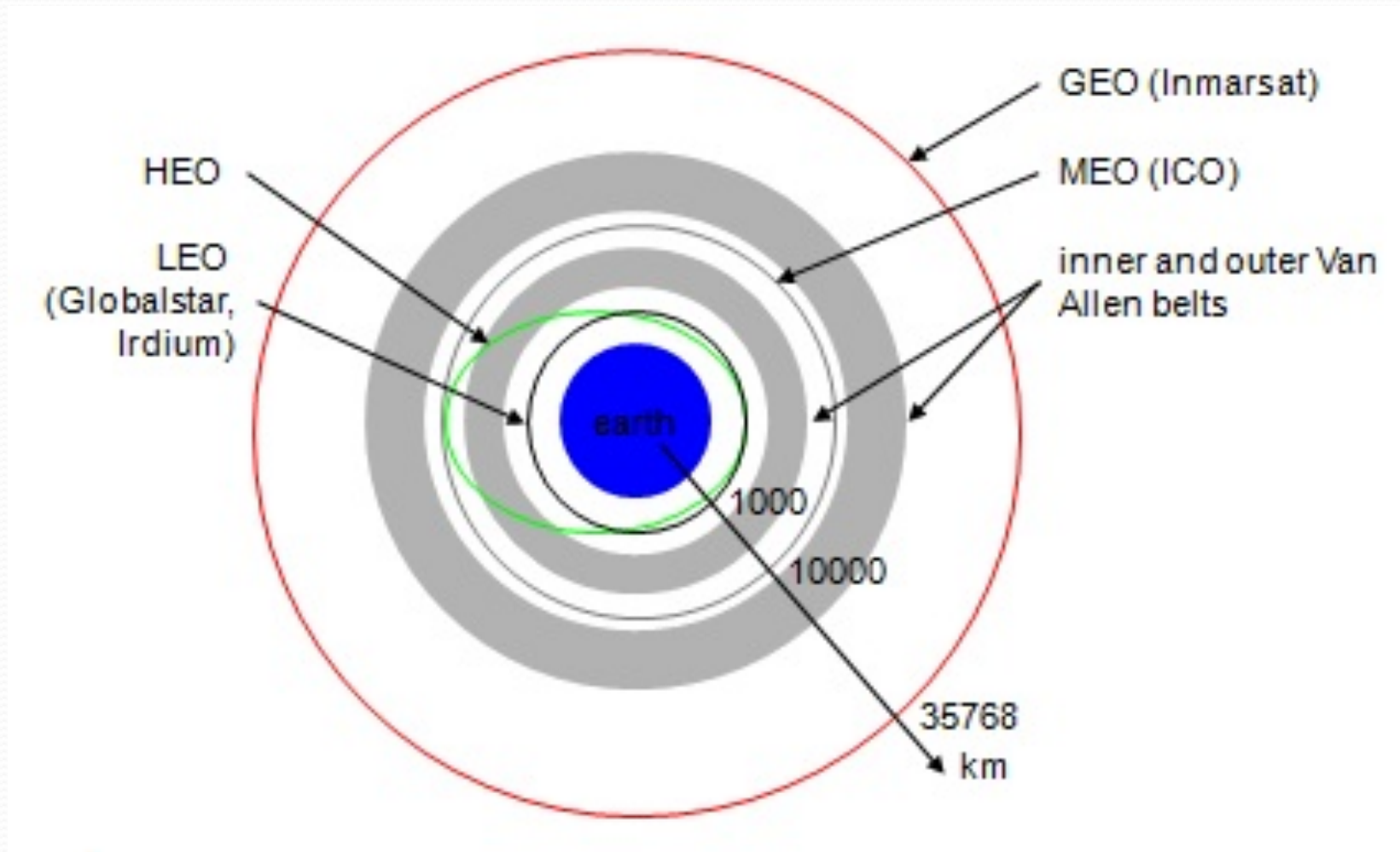
Example: satellite systems at 4-6 GHz



Satellite Orbits

- Four different types of satellite orbits can be identified depending on the shape and diameter of the orbit:
 - **Geostationary Earth Orbit (GEO):** 36000 km above earth surface
 - Most of TV & radio broadcast satellites
 - **Medium Earth Orbit (MEO):** 5000-12,000 km
 - Intermediate Circular Orbit (ICO)
 - **Low Earth Orbit (LEO):** 500-1500 km
 - **Highly Elliptical Orbit (HEO):** satellites with non-circular orbits.

Satellite Orbits



Geostationary satellites (GEO)

- Orbit 35.786 km distance to earth surface, orbit in equatorial plane (inclination 0°)
 - Complete rotation exactly one day, **satellite is synchronous to earth rotation**
 - **Fix antenna positions**, no adjusting necessary
 - Satellites typically have a **large footprint** (cover bigger area)
 - **High transmit power** needed
 - **High latency** due to long distance
 - Not useful for global coverage for small mobile phones and data transmission, typically **used for radio and TV transmission**

Low Earth Orbit (LEO)

- **Global radio coverage** possible
- **Smaller footprints**, better frequency reuse
- **Handover necessary** from one satellite to another
- **Many satellites necessary for global coverage**
- **More complex** systems due to moving satellites

Medium Earth Orbit (MEO)

- **Slower** moving satellites
- **Less satellites** needed
- **Simpler** system design
- For many connections **no hand-over** needed
- **Higher latency**
- **Higher sending power** needed
- **Special antennas** for small footprints needed

Routing

- **Satellite Routing:** Routing of data transmission from one user to another.
 - Two strategies:
 - **Between Inter satellite link (ISL)**
 - Traffic is routed **between the satellites**
 - **Advantage:** only one uplink and one downlink , offers lower latency
 - **Disadvantage: system complexity** due to additional antennas
 - **Relayed through earth station**
 - Traffic is routed between earth station and satellite
 - **Disadvantage:** two uplinks and two downlinks needed

Localization

- **Satellite gateways:** maintains several **registers**
- **HLR:** stores all **static information about a user** as well as his/her **current location**
- **VLR:** maintains the **last known location of a mobile user**
- **Satellite User Mapping Register(SUMR):** Stores the **current position of satellites** and a mapping of each **user to the current satellite**
- **Registration:** mobile station sends a signal which may be received by one or several satellites
 - Satellites receiving such signal report this event to gateway
 - Gateway determines the location of the user via the location of the satellites
 - User data is requested from the user's HLR, VLR & SUMR are updated

Handover

- Handover in satellite systems caused by the movement of satellites
- Four types
 - **Intra-satellite handover**
 - Handover from **one spot beam to another**
 - Mobile station still in the footprint of the satellite, but in another cell
 - **Inter-satellite handover**
 - Handover from one **satellite to another satellite**
 - Mobile station leaves the footprint of one satellite
 - **Gateway handover**
 - Handover from **one gateway to another**
 - Mobile station still in the footprint of a satellite, but gateway leaves the footprint
 - **Inter-system handover**
 - Handover from the **satellite network to a terrestrial cellular network**
 - Mobile station can reach a terrestrial network again which might be cheaper, has a lower latency etc